

Simple Linear Regression

Estimation by Ordinary Least Squares

Tu Li

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Econ 453 University of Arizona

Roadmap

1. From Statistical Learning to a Linear Model
2. The Simple Linear Regression Model
3. Fitting the Line — Residuals and RSS
4. The OLS Solution
5. From Pencil to Software
6. OLS as Loss Minimization
7. Summary

1. From Statistical Learning to a Linear Model

1.1 Where We Left Off

Recall the framework from Lecture 5:

$$Y = f(X) + \epsilon, \quad \mathbb{E}[\epsilon | X] = 0.$$

Two questions were left open:

- **What is f ?** In Lecture 2 §10 we identified the ideal f as the conditional expectation $f(x) = \mathbb{E}[Y | X = x]$, but that's an abstract object — we cannot compute it from data without further assumptions.
- **How do we estimate f from data?** A general strategy needs some structure on f to make estimation possible.

This lecture takes the simplest possible structure — f is a straight line — and shows how to estimate it.

1.2 The Linear Restriction

We assume the regression function is **linear** in X :

$$f(X) = \beta_0 + \beta_1 X.$$

Plugging into $Y = f(X) + \epsilon$ gives the **simple linear regression model**:

$$Y = \beta_0 + \beta_1 X + \epsilon.$$

- β_0 is the **intercept** — the value of f at $X = 0$.
- β_1 is the **slope** — the change in f for a one-unit increase in X .
- Both are **unknown constants** that we must estimate from data.

Convention. Greek letters $(\alpha, \beta, \gamma, \dots)$ denote unknown population parameters throughout this course.

1.3 The Advertising Motivation

ISLR's running example: a company sells a product in 200 markets. For each market we know spending on TV, radio, and newspaper advertising, plus the observed **sales**. As a consultant, we want to use this data to understand and predict next year's sales.

Questions a linear regression can address:

- Is there a relationship between advertising and sales?
- How strong is it?
- Which medium is most strongly associated with sales?
- How accurately can we predict sales from spending levels?
- Is the relationship really linear?
- Are there interactions between media?

We'll work with the simplest version — one predictor, one response (say sales on TV) — in this lecture. Multiple predictors come in Lecture 9.

2. The Simple Linear Regression Model

2.1 The Model

Definition — Simple Linear Regression

For random variables X (predictor) and Y (response),

$$Y \approx \beta_0 + \beta_1 X \quad (\text{approximate, systematic})$$

is made exact by adding a mean-zero error:

$$Y = \beta_0 + \beta_1 X + \epsilon, \quad \mathbb{E}[\epsilon \mid X] = 0.$$

We say we are **regressing Y on X** . In the advertising example, with $Y =$ sales and $X =$ TV spend,

$$\text{sales} = \beta_0 + \beta_1 \cdot \text{TV} + \epsilon.$$

2.2 The Linear Function f

The systematic piece $f(x) = \beta_0 + \beta_1 x$ is a **function** in the formal sense:

$$f : \mathbb{R} \longrightarrow \mathbb{R}, \quad x \longmapsto \beta_0 + \beta_1 x.$$

Notation reminder. A function is a rule that takes each input (here, a real number x) and returns exactly one output (here, the real number $\beta_0 + \beta_1 x$). Many objects in this course are functions: random variables, PDFs, the regression function, the loss function we are about to define.

- **Intercept** β_0 : the value $f(0)$; where the line crosses the vertical axis.
- **Slope** β_1 : how much $f(x)$ changes when x increases by one unit.
Positive slope $\Rightarrow$ line goes up; negative slope $\Rightarrow$ line goes down.
- **Key property**: the change in $f(x)$ is *proportional* to the change in x .
The graph is a straight line.

2.3 Random Variables vs. Realizations

The model is a statement about random variables (X, Y) . The data contain n observed **realizations** (x_i, y_i) . Each one satisfies

$$y_i = \beta_0 + \beta_1 x_i + \epsilon_i, \quad i = 1, \dots, n,$$

where ϵ_i is the unobservable noise affecting observation i .

We treat the pairs (X_i, Y_i) as **independent across i** . Separately, $\mathbb{E}[\epsilon_i | X_i] = 0$ within each pair: the error has no systematic direction at any predictor value.

Notation reminder. Uppercase (X, Y) for random variables; lowercase (x_i, y_i) for the numbers we actually see. Same convention as Lectures 1–4.

2.4 Why Include an Intercept?

The model is sometimes written without β_0 ,

$$Y = \beta_1 X + \epsilon \quad (\text{no intercept}),$$

but this is a strong assumption: it forces the line to pass through the origin.

Setting $\beta_0 = 0$ means

$$\mathbb{E}[Y \mid X = 0] = \beta_0 = 0.$$

In words: **when X is zero, the expected value of Y is zero too**. For most problems this is not realistic — a market with zero TV spend still sells some product.

Forcing $\beta_0 = 0$ when the truth is nonzero distorts the slope and gives worse predictions. **We always include β_0 unless there is a specific reason not to.**

3. Fitting the Line — Residuals and RSS

3.1 What We Want to Compute

Since β_0 and β_1 are unknown, our job is to produce **estimates** from the data. We denote these

$$\hat{\beta}_0, \hat{\beta}_1.$$

The hats are a reminder that these are computed from a noisy sample, so in general $\hat{\beta}_0 \neq \beta_0$ and $\hat{\beta}_1 \neq \beta_1$ — we have **estimation error**.

Given any choice of $\hat{\beta}_0, \hat{\beta}_1$, the **fitted value** for observation i is

$$\hat{y}_i = \hat{\beta}_0 + \hat{\beta}_1 x_i,$$

and the **prediction** for a new input x_{new} is

$$\hat{y}_{\text{new}} = \hat{\beta}_0 + \hat{\beta}_1 x_{\text{new}}.$$

We still have to pick *which* $(\hat{\beta}_0, \hat{\beta}_1)$ to use. The next two frames build that criterion.

3.2 Residuals

For each observation i , the gap between the observed y_i and the fitted $\hat{y}_i$ is the **residual**:

$$e_i = y_i - \hat{y}_i = y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i.$$

- $e_i > 0$: the model *under*-predicts observation i .
- $e_i < 0$: the model *over*-predicts observation i .
- $e_i = 0$: the line passes exactly through the data point.

Good estimates should make the residuals small **on average**. But we need to be careful what “small” means — positive and negative residuals would just cancel if we summed them directly.

3.3 Residual Sum of Squares

To prevent cancellation, we **square** each residual before summing.

Definition — Residual Sum of Squares

$$\text{RSS} = e_1^2 + e_2^2 + \cdots + e_n^2 = \sum_{i=1}^n e_i^2.$$

Substituting $e_i = y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i$,

$$\text{RSS}(\hat{\beta}_0, \hat{\beta}_1) = \sum_{i=1}^n (y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i)^2.$$

Two key features:

- Every term is nonnegative; $\text{RSS} \geq 0$.
- $\text{RSS} = 0$ would require every point to lie exactly on the line — essentially never happens with real data.

3.4 Ordinary Least Squares

Key Result — The OLS criterion

The **ordinary least squares** (OLS) estimates are the pair $(\hat{\beta}_0, \hat{\beta}_1)$ that minimize the RSS:

$$(\hat{\beta}_0, \hat{\beta}_1) = \arg \min_{\beta_0, \beta_1} \sum_{i=1}^n (y_i - \beta_0 - \beta_1 x_i)^2.$$

This single idea — pick parameters to minimize a sum of squared errors — is the template for almost every ML method we will see.

3.5 OLS as the Prototype

Many methods later in the course follow the same three-step recipe:

1. Pick a model family ($\beta_0 + \beta_1 X$ here).
2. Define a **loss** on the data (RSS here).
3. Minimize the loss with respect to the parameters.

Where this recipe reappears

- **Multiple regression** (L9): same RSS, more β_j 's.
- **Ridge / lasso** (L10): RSS *plus* a penalty $\lambda \|\beta\|_2^2$ or $\lambda \|\beta\|_1$.
- **Logistic regression** (L12): swap RSS for the log-likelihood loss; still minimize via calculus.
- **Resampling, trees, random forests** (L13–L14): different loss functions, same minimize-by-calculus or minimize-by-search principle.

3.6 OLS Has a Closed-Form Solution

We have written down an optimization problem,

$$\min_{\beta_0, \beta_1} \sum_{i=1}^n (y_i - \beta_0 - \beta_1 x_i)^2.$$

This problem has a clean closed-form solution — a pair of formulas for $(\hat{\beta}_0, \hat{\beta}_1)$ that depend only on the data.

Solving it requires calculus: partial derivatives in β_0 and β_1 , convex optimization to know the critical point is the global minimum, and a few lines of algebra.

Strategy of this lecture. We separate the *solution* from the *derivation*.

- Section 4 states the OLS formulas and explores their consequences.
- The **Appendix** at the end (Appendix A–D) contains the calculus toolkit and the line-by-line derivation, for students who want to see *why* the formulas take the form they do.

Onward to §4: the OLS solution.

4. The OLS Solution

4.1 The OLS Estimators

Key Result — OLS estimators (one predictor)

$$\hat{\beta}_1 = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^n (x_i - \bar{x})^2}, \quad \hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}.$$

- The numerator of $\hat{\beta}_1$ is the sample covariance between x and y (up to dividing by $n - 1$); the denominator is the sample variance of x . So $\hat{\beta}_1$ scales how much y and x **co-vary** by how spread out x is.
- $\hat{\beta}_0$ is whatever shift makes the line pass through the sample mean point $(\bar{x}, \bar{y})$.

This pair of formulas is the entire content of OLS in one predictor. From here, everything is *using* them. (Derivation: Appendix D.)

4.2 When Is $\hat{\beta}_1$ Undefined?

Look at the denominator: $\sum_i (x_i - \bar{x})^2 = 0$ exactly when every x_i equals $\bar{x}$ — in words, when the predictor is constant across the sample.

- No variation in $X \Rightarrow$ we cannot tell from the data how Y changes with X .
- Computation will return an error or a warning when this happens.
- In practice this is rare with continuous predictors but occasionally arises with binary or categorical inputs that happen to be all the same in a small sample.

Forward pointer. The same issue, generalized to many predictors, becomes **multicollinearity**: Lecture 9 exposes the issue, and Lecture 10 addresses it with ridge regression.

4.3 A Hidden Statistical Identity

Look again at the slope estimator:

$$\hat{\beta}_1 = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^n (x_i - \bar{x})^2}.$$

Multiply numerator and denominator by $\frac{1}{n}$ (the ratio is unchanged):

$$\hat{\beta}_1 = \frac{\frac{1}{n} \sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\frac{1}{n} \sum_{i=1}^n (x_i - \bar{x})^2}.$$

Now name the two pieces:

- **Denominator:** $\frac{1}{n} \sum_i (x_i - \bar{x})^2$ is the **sample variance** of X (up to the n vs. $n - 1$ convention; Lecture 3 §7.1).
- **Numerator:** $\frac{1}{n} \sum_i (x_i - \bar{x})(y_i - \bar{y})$ is the **sample covariance** of X and Y , the sample analogue of covariance from Lecture 2 §5.4.

So the OLS slope is literally “sample covariance over sample variance.”

What does that converge to?

4.4 Applying the Law of Large Numbers

Recall — LLN (Lecture 3 §5.1)

If $W_1, W_2, \dots$ are i.i.d. with $\mathbb{E}[|W|] < \infty$,

$$\frac{1}{n} \sum_{i=1}^n W_i \xrightarrow{P} \mathbb{E}[W] \quad \text{as } n \rightarrow \infty.$$

Apply this to the two sums in $\hat{\beta}_1$, taking $W = (X - \mathbb{E}[X])^2$ for the denominator and $W = (X - \mathbb{E}[X])(Y - \mathbb{E}[Y])$ for the numerator:

$$\frac{1}{n} \sum_i (x_i - \bar{x})^2 \xrightarrow{P} \text{Var}(X), \quad \frac{1}{n} \sum_i (x_i - \bar{x})(y_i - \bar{y}) \xrightarrow{P} \text{Cov}(X, Y).$$

Aside. The centering at $\bar{x}$ rather than at $\mathbb{E}[X]$ is harmless in the limit: $\bar{x} \xrightarrow{P} \mathbb{E}[X]$ by LLN, and the ratio rule for limits closes the loop.

Key Result — Statistical limit of $\hat{\beta}_1$

$$\hat{\beta}_1 \xrightarrow{P} \frac{\text{Cov}(X, Y)}{\text{Var}(X)}.$$

4.5 Geometry Meets Statistics

Why this is remarkable

1. We set up OLS as a **pure geometry** problem: find the line that makes the total squared vertical gap as small as possible. *No probability appeared in the derivation — only calculus and convexity.*
2. Yet the answer, in large samples, is a **purely statistical** object:
$$\frac{\text{Cov}(X, Y)}{\text{Var}(X)}$$
 — a ratio of moments of the joint distribution of (X, Y) .
3. Geometry and statistics gave the same answer. **I find this deeply amazing.**

5. From Pencil to Software

5.1 Computation Moves to the Computer

We have closed-form formulas for $\hat{\beta}_0$ and $\hat{\beta}_1$. In principle we could compute them by hand.

In practice, real datasets have n in the thousands or millions — far too many sums to do with pencil and paper. Starting in Lecture 7 we hand the arithmetic to a computer.

The formulas matter for understanding. In practice, computer software handles the arithmetic for us.

6. OLS as Loss Minimization

6.1 The Recipe and What's Next

The loss-minimization template

1. Choose a model family. Here: $f(X; \beta_0, \beta_1) = \beta_0 + \beta_1 X$.
2. Choose a loss. Here: $\text{RSS}(\beta_0, \beta_1)$.
3. Minimize. Here: set both partials to zero (Appendix D).

Every parametric method from L9 onward changes one of these choices — never the recipe.

We now have formulas that produce numbers, but several questions remain open:

- Sampling variability of $\hat{\beta}_0, \hat{\beta}_1$ across samples.
- Bias — are they centered on the true β_0, β_1 ?
- Inference — CIs and p -values for $\hat{\beta}_1$.
- Goodness of fit — how much of $\text{Var}(Y)$ does the line explain?
(Lecture 9.)

Lecture 7 answers the first three using the L1–L4 toolkit; Lecture 9 develops goodness-of-fit measures.

7. Summary

7.1 Take Away — The Model

Simple linear regression in one slide

- **Model within one observation:** $Y = \beta_0 + \beta_1 X + \epsilon$ with $\mathbb{E}[\epsilon \mid X] = 0$.
- **Data:** n independent realized pairs (x_i, y_i) , $i = 1, \dots, n$.
- **Fitted value:** $\hat{y}_i = \hat{\beta}_0 + \hat{\beta}_1 x_i$.
- **Residual:** $e_i = y_i - \hat{y}_i$.
- **Loss:** $\text{RSS} = \sum_i e_i^2$.

7.2 Take Away — The Estimators

OLS estimators (derived in Appendix D)

$$\hat{\beta}_1 = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^n (x_i - \bar{x})^2}, \quad \hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}.$$

Obtained by minimizing $\text{RSS}(\beta_0, \beta_1)$ — a convex differentiable function — via the first-order condition: set $\partial \text{RSS} / \partial \beta_0 = 0$ and $\partial \text{RSS} / \partial \beta_1 = 0$.

In the limit, the LLN gives the statistical identity

$$\hat{\beta}_1 \xrightarrow{p} \frac{\text{Cov}(X, Y)}{\text{Var}(X)} \quad (\text{geometry meets statistics, §4.3–§4.5}).$$

7.3 Where Each Tool Reappears

Forward pointers

- **Inference** — Lecture 7. Standard errors, CIs, and large-sample coefficient tests for $\hat{\beta}_0, \hat{\beta}_1$, using the L1–L4 toolkit.
- **Matrix form** — Lecture 8 repackages OLS to scale painlessly to many predictors.
- **Multiple regression** — Lecture 9. Same recipe, p predictors.
- **Ridge / lasso** — Lecture 10. RSS plus a penalty; same minimization.
- **Logistic regression** — Lecture 12. Different loss, same calculus strategy.
- **Calculus toolkit (Appendix A–C)** — the same machinery is reused in every parametric method from here on.

Next time: how trustworthy are $\hat{\beta}_0$ and $\hat{\beta}_1$? Inference for the simple linear regression.

Appendix A. Calculus Toolkit — The Derivative

A.1 Definition of the Derivative

Definition — Derivative

Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be a real-valued function. The **derivative** of f at a point x , written $f'(x)$ or $\frac{df}{dx}$, is the limit

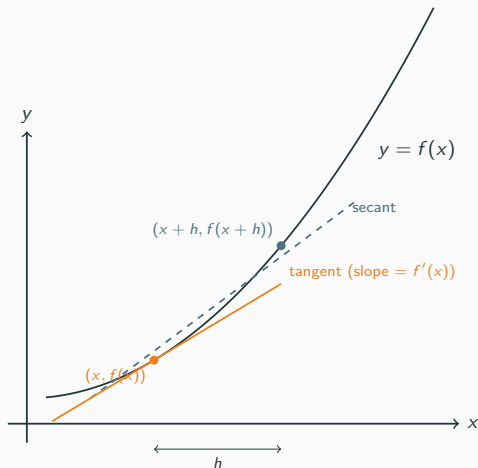
$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h},$$

provided this limit exists.

The quotient $\frac{f(x+h) - f(x)}{h}$ is the slope of the **secant line** connecting two nearby points on the graph. Letting $h \rightarrow 0$ collapses the secant to a line that just *touches* the graph at x — the **tangent line**.

Reading. “ f prime of x ” or “ df over dx .”

A.2 Picture: Secant Collapsing to Tangent



As $h \rightarrow 0$ the secant (blue dashed) pivots and approaches the tangent (orange solid). The slope of that limiting tangent is the derivative $f'(x)$.

A.3 Example: Derivative of $f(x) = x^2$ from the Definition

Setup. Let $f(x) = x^2$. Compute $f'(x)$ using the limit.

Step 1. Form the difference quotient:

$$\frac{f(x+h) - f(x)}{h} = \frac{(x+h)^2 - x^2}{h}.$$

Step 2. Expand and simplify:

$$(x+h)^2 - x^2 = x^2 + 2xh + h^2 - x^2 = 2xh + h^2.$$

Divide by h :

$$\frac{2xh + h^2}{h} = 2x + h.$$

Step 3. Take $h \rightarrow 0$:

$$f'(x) = \lim_{h \rightarrow 0} (2x + h) = 2x.$$

Interpretation. At any point x , the parabola $y = x^2$ has tangent slope $2x$. At $x = 3$ the curve is rising with slope 6; at $x = 0$ the tangent is horizontal (slope 0) — which is where the parabola bottoms out.

A.4 Theory and Intuition

Theory. The derivative $f'(x)$ measures how fast f changes per unit change in x at the point x . It is the slope of the tangent line — the best linear approximation to f near x . Formally,

$$f(x + h) \approx f(x) + f'(x) h \quad \text{for small } h.$$

Intuition. Zoom in close enough to a smooth curve and it starts to look like a straight line. The slope of that straight line is the derivative. Three readings to keep in mind:

- **Geometric:** slope of the tangent.
- **Rate-of-change:** how much $f(x)$ moves when you nudge x a little.
- **Linear approximation:** the coefficient on h in $f(x + h) \approx f(x) + f'(x) h$.

All three readings agree — and all three matter for what's coming. The minimum of f is where the slope is zero (geometric); small parameter changes barely move the loss (rate); and gradient descent is exactly the linear-approximation reading.

Appendix B. Calculus Toolkit — Rules of Differentiation

B.1 Why We Need Rules

Computing $f'(x)$ from the limit definition every time is painful — you saw the algebra even for x^2 . Fortunately, derivatives obey a handful of **algebraic rules** that let us differentiate any expression made from sums, products, powers, and compositions without going back to the limit.

We will need four rules to handle RSS:

1. **Power rule** — to differentiate x^k .
2. **Sum rule** — to handle the $\sum_i$ in RSS.
3. **Scalar-multiple rule** — to pull out constants.
4. **Chain rule** — to differentiate (something)².

And one extension for multivariable functions: **partial derivatives**.

B.2 Power, Sum, and Scalar-Multiple Rules

Rule 1 — Power

For any real number k ,

$$\frac{d}{dx} x^k = k x^{k-1}.$$

Rule 2 — Sum

$$\frac{d}{dx} (f(x) + g(x)) = f'(x) + g'(x).$$

Rule 3 — Scalar Multiple

For any constant c ,

$$\frac{d}{dx} (c f(x)) = c f'(x).$$

Examples. $\frac{d}{dx} x^5 = 5x^4.$ $\frac{d}{dx} x^{-2} = -2x^{-3}.$ $\frac{d}{dx} (x^3 + 5x) = 3x^2 + 5.$
 $\frac{d}{dx} (7x^4) = 28x^3.$

B.3 The Chain Rule

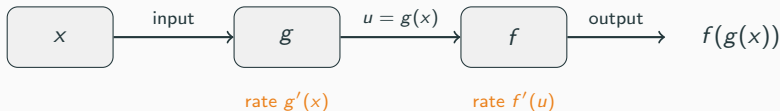
Rule 4 — Chain Rule

If $u = g(x)$ and f is a function of u ,

$$\frac{d}{dx}f(g(x)) = f'(g(x)) \cdot g'(x).$$

“Differentiate the outside, evaluated at the inside, times the derivative of the inside.”

Picture of the composition $f \circ g$:



Rates multiply: change x a little, g moves at rate $g'(x)$, then f moves at rate $f'(g(x))$ relative to its input. Total rate: $f'(g(x)) \cdot g'(x)$.

B.4 Chain Rule — Worked Examples

Example 1. $\frac{d}{dx}(3x + 1)^2$.

Outside: u^2 , derivative $2u$. Inside: $u = 3x + 1$, derivative 3. So

$$\frac{d}{dx}(3x + 1)^2 = 2(3x + 1) \cdot 3 = 6(3x + 1).$$

Example 2. $\frac{d}{dx}\sqrt{2x^2 + 1}$.

Rewrite: $\sqrt{u} = u^{1/2}$, derivative $\frac{1}{2}u^{-1/2}$. Inside: $u = 2x^2 + 1$, derivative $4x$. So

$$\frac{d}{dx}\sqrt{2x^2 + 1} = \frac{1}{2}(2x^2 + 1)^{-1/2} \cdot 4x = \frac{2x}{\sqrt{2x^2 + 1}}.$$

Example 3 (preview of OLS). $\frac{\partial}{\partial \beta_0}(y_i - \beta_0 - \beta_1 x_i)^2$.

Outside: u^2 , derivative $2u$. Inside (in β_0): $u = y_i - \beta_0 - \beta_1 x_i$, derivative -1 . So

$$\frac{\partial}{\partial \beta_0}(y_i - \beta_0 - \beta_1 x_i)^2 = 2(y_i - \beta_0 - \beta_1 x_i) \cdot (-1).$$

This is the workhorse calculation behind OLS.

B.5 Partial Derivatives

Definition — Partial Derivative

For a function of two (or more) variables $f(x, y)$, the **partial derivative of f with respect to x** is

$$\frac{\partial f}{\partial x}(x, y) = \lim_{h \rightarrow 0} \frac{f(x + h, y) - f(x, y)}{h}.$$

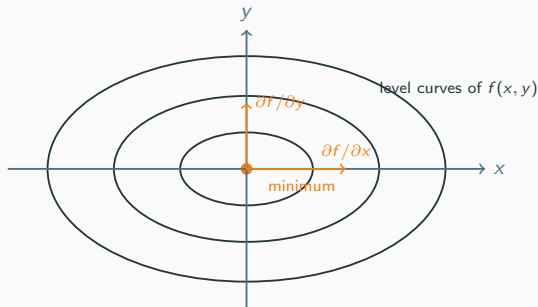
Hold y fixed and differentiate as if f were a function of x alone. Similarly for $\partial f / \partial y$.

Example. $f(x, y) = x^2y + 3y$.

$$\frac{\partial f}{\partial x} = 2xy, \quad \frac{\partial f}{\partial y} = x^2 + 3.$$

Intuition. A surface over the (x, y) plane has many “slopes” at a point, one for each direction. $\partial f / \partial x$ is the slope you’d feel walking strictly east; $\partial f / \partial y$ is the slope walking strictly north.

B.6 Picture: Slopes in Two Directions



Each ellipse is a level set: along it f is constant. The two partial derivatives are the slopes felt as you step strictly along the x -axis or strictly along the y -axis from a point. At the minimum (orange dot) the surface is flat in every direction, so **both partials are zero**.

Forward pointer. The OLS problem has two parameters, β_0 and β_1 , so we will set **both partials** $\partial \text{RSS} / \partial \beta_0$ and $\partial \text{RSS} / \partial \beta_1$ to zero in Appendix D.

Appendix C. Calculus Toolkit — Convex Optimization

C.1 Convex Functions

Definition — Convex Function

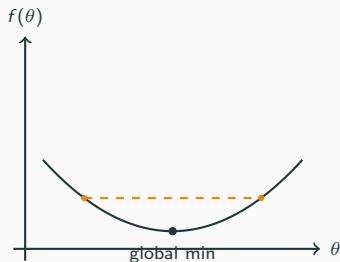
A function $f : \mathbb{R}^k \rightarrow \mathbb{R}$ is **convex** if for every θ_1, θ_2 and every $t \in [0, 1]$,

$$f(t\theta_1 + (1 - t)\theta_2) \leq tf(\theta_1) + (1 - t)f(\theta_2).$$

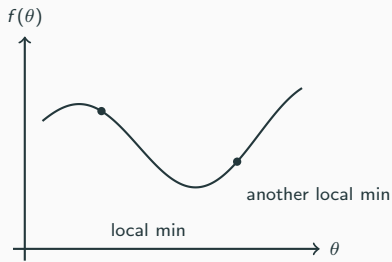
Intuition. The chord connecting two points on the graph of f lies *on or above* the graph itself. Convex functions are “bowl-shaped” — they curve upward.

Theory. A convex function has at most *one* minimum value, and any point where the function is locally as low as possible is in fact the global minimum. Non-convex functions can have multiple “valleys,” which is what makes them hard to optimize.

C.2 Picture: Convex vs Non-Convex



Convex
one bottom



Non-convex
many bottoms

Convex (left): the dashed chord between any two points sits above the curve. Non-convex (right): different starting points lead to different “minima” — you cannot tell which is best without searching everywhere.

C.3 First-Order Condition for a Minimum

Key Result — Optimality Condition

Let $f : \mathbb{R}^k \rightarrow \mathbb{R}$ be differentiable and convex, with parameter vector $\theta = (\theta_1, \dots, \theta_k)$. Then θ^* minimizes f if and only if every partial derivative vanishes there:

$$\frac{\partial f}{\partial \theta_1}(\theta^*) = \frac{\partial f}{\partial \theta_2}(\theta^*) = \dots = \frac{\partial f}{\partial \theta_k}(\theta^*) = 0.$$

Intuition. At the bottom of a bowl, the surface is *flat in every direction*. Tilting any partial-derivative “compass needle” to zero is what nails the bottom down.

Non-differentiable convex functions need a small generalization (subgradients), and non-convex functions need fancier tools. Our RSS is differentiable *and* convex, so this result is exactly what we need.

C.4 RSS Is Convex

Why does the optimality condition apply to RSS? Because RSS is a sum of squared linear functions of (β_0, β_1) :

$$\text{RSS}(\beta_0, \beta_1) = \sum_{i=1}^n (y_i - \beta_0 - \beta_1 x_i)^2.$$

- Each term $(y_i - \beta_0 - \beta_1 x_i)^2$ is a squared linear function of (β_0, β_1) , which is convex.
- Sums of convex functions are convex.

Therefore RSS is convex in (β_0, β_1) — exactly the regularity needed for the first-order condition of Appendix C.3 to apply.

Setting both partial derivatives to zero will therefore find *the* minimum, not merely a critical point. That is exactly what OLS does, and we carry out the calculation in Appendix D.

Appendix D. Deriving the OLS Estimators

D.1 The Plan

We want the pair $(\hat{\beta}_0, \hat{\beta}_1)$ minimizing

$$\text{RSS}(\beta_0, \beta_1) = \sum_{i=1}^n (y_i - \beta_0 - \beta_1 x_i)^2.$$

RSS is differentiable and convex (Appendix C.4), so the optimality condition (Appendix C.3) says: **set both partial derivatives to zero.**

1. Compute $\partial \text{RSS} / \partial \beta_0$. Set to zero $\Rightarrow$ **normal equation 1.**
2. Compute $\partial \text{RSS} / \partial \beta_1$. Set to zero $\Rightarrow$ **normal equation 2.**
3. Solve the two normal equations for $(\hat{\beta}_0, \hat{\beta}_1)$.

Each step uses tools from Appendix A–B only — no new ideas.

D.2 Partial with Respect to β_0

Differentiate under the sum (sum rule). Inside each squared term, use the chain rule with outside u^2 and inside $u = y_i - \beta_0 - \beta_1 x_i$ (so $\partial u / \partial \beta_0 = -1$):

$$\begin{aligned}\frac{\partial \text{RSS}}{\partial \beta_0} &= \sum_{i=1}^n \frac{\partial}{\partial \beta_0} (y_i - \beta_0 - \beta_1 x_i)^2 \\&= \sum_{i=1}^n 2(y_i - \beta_0 - \beta_1 x_i) \cdot \frac{\partial}{\partial \beta_0} (y_i - \beta_0 - \beta_1 x_i) \\&= \sum_{i=1}^n 2(y_i - \beta_0 - \beta_1 x_i) \cdot (-1) \\&= -2 \sum_{i=1}^n (y_i - \beta_0 - \beta_1 x_i).\end{aligned}$$

D.3 Normal Equation 1, and $\hat{\beta}_0$

Set the partial derivative to zero at $(\hat{\beta}_0, \hat{\beta}_1)$:

$$\begin{aligned} -2 \sum_{i=1}^n (y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i) &= 0 \\ \sum_{i=1}^n y_i - n\hat{\beta}_0 - \hat{\beta}_1 \sum_{i=1}^n x_i &= 0. \end{aligned}$$

This is the **first normal equation**. Divide both sides by n , using the sample means $\bar{y} = \frac{1}{n} \sum_i y_i$ and $\bar{x} = \frac{1}{n} \sum_i x_i$:

$$\bar{y} - \hat{\beta}_0 - \hat{\beta}_1 \bar{x} = 0.$$

Key Result — Intercept formula

$$\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}.$$

This tells us $\hat{\beta}_0$ *in terms of* $\hat{\beta}_1$ — we still need a second equation to pin $\hat{\beta}_1$ down.

D.4 Partial with Respect to β_1

Same logic, but the inside derivative changes: $\partial u / \partial \beta_1 = -x_i$. So

$$\begin{aligned}\frac{\partial \text{RSS}}{\partial \beta_1} &= \sum_{i=1}^n \frac{\partial}{\partial \beta_1} (y_i - \beta_0 - \beta_1 x_i)^2 \\ &= \sum_{i=1}^n 2(y_i - \beta_0 - \beta_1 x_i) \cdot (-x_i) \\ &= -2 \sum_{i=1}^n x_i (y_i - \beta_0 - \beta_1 x_i).\end{aligned}$$

D.5 Normal Equation 2

Set it to zero at $(\hat{\beta}_0, \hat{\beta}_1)$ and expand:

$$\begin{aligned} -2 \sum_{i=1}^n x_i (y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i) &= 0 \\ \sum_{i=1}^n x_i y_i - \hat{\beta}_0 \sum_{i=1}^n x_i - \hat{\beta}_1 \sum_{i=1}^n x_i^2 &= 0. \end{aligned}$$

Rearranged, the **second normal equation** is

$$\hat{\beta}_0 \sum_{i=1}^n x_i + \hat{\beta}_1 \sum_{i=1}^n x_i^2 = \sum_{i=1}^n x_i y_i.$$

We now have two equations in two unknowns. Substituting the intercept formula from D.3 will solve for $\hat{\beta}_1$.

D.6 Solving for $\hat{\beta}_1$

Substitute $\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}$ into the second normal equation, and use $\sum_i x_i = n\bar{x}$:

$$\begin{aligned}(\bar{y} - \hat{\beta}_1 \bar{x}) n\bar{x} + \hat{\beta}_1 \sum_i x_i^2 &= \sum_i x_i y_i \\ n\bar{x} \bar{y} - \hat{\beta}_1 n\bar{x}^2 + \hat{\beta}_1 \sum_i x_i^2 &= \sum_i x_i y_i \\ \hat{\beta}_1 \left(\sum_i x_i^2 - n\bar{x}^2 \right) &= \sum_i x_i y_i - n\bar{x} \bar{y}.\end{aligned}$$

Two algebraic identities (proof on next frame as a sidebar):

$$\sum_i x_i^2 - n\bar{x}^2 = \sum_i (x_i - \bar{x})^2, \quad \sum_i x_i y_i - n\bar{x} \bar{y} = \sum_i (x_i - \bar{x})(y_i - \bar{y}).$$

Therefore

$$\hat{\beta}_1 = \frac{\sum_i x_i y_i - n\bar{x} \bar{y}}{\sum_i x_i^2 - n\bar{x}^2} = \frac{\sum_i (x_i - \bar{x})(y_i - \bar{y})}{\sum_i (x_i - \bar{x})^2}.$$

D.7 Sidebar: The Two Identities

Both identities follow from expanding the square / product and using $\sum_i x_i = n\bar{x}$, $\sum_i y_i = n\bar{y}$.

Identity 1.

$$\begin{aligned}\sum_i (x_i - \bar{x})^2 &= \sum_i (x_i^2 - 2x_i\bar{x} + \bar{x}^2) \\ &= \sum_i x_i^2 - 2\bar{x} \sum_i x_i + n\bar{x}^2 \\ &= \sum_i x_i^2 - 2\bar{x} \cdot n\bar{x} + n\bar{x}^2 = \sum_i x_i^2 - n\bar{x}^2.\end{aligned}$$

Identity 2.

$$\begin{aligned}\sum_i (x_i - \bar{x})(y_i - \bar{y}) &= \sum_i (x_i y_i - \bar{y} x_i - \bar{x} y_i + \bar{x} \bar{y}) \\ &= \sum_i x_i y_i - \bar{y} \cdot n\bar{x} - \bar{x} \cdot n\bar{y} + n\bar{x} \bar{y} \\ &= \sum_i x_i y_i - n\bar{x} \bar{y}.\end{aligned}$$